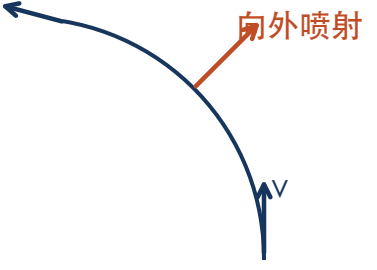

$$m\gamma_v v\, d\theta = u\, dm_0$$

$$m_0/m_1 = \exp(\gamma_v v|\theta|/u)$$